Adrenal Glands

The adrenal glands, which are in the abdominal area, are located bilaterally just above each kidney and the diaphragm. The adrenal glands consist of two layers, the cortex (outer layer) and the medulla (inner core).

The cortex secretes:

- Androgen - male hormone
- Aldosterone - controls blood pressure and fluid balance
- Cortisol - regulates and coordinates metabolism

The medulla secretes stress reaction hormones such as:

- Adrenaline
- Noradrenaline
- Catecholamine

The Pancreas

The pancreas is located behind the stomach. The islets of Langerhans secrete glucagon, insulin, pancreatic polypeptide and somatostatin. The pancreas produces and releases digestive enzymes and juices that break down foods as they enter the small intestine.

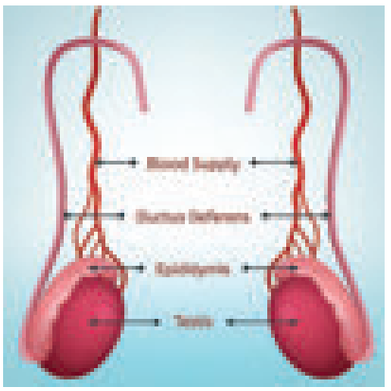
NOTE: Pancreas is called as a mixed gland since it serves both as an endocrine gland and as a digestive organ and exocrine body.

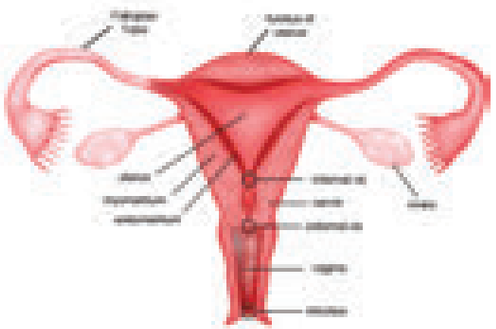
Testes and Ovaries

Ovaries and Testes are the endocrine glands and gonads, which is called as a sex and reproduction glandular structure.

The ovaries produce progesterone, estrogen, inhibin and androstenedione. Progesterone regulates menstrual cycle and the preparation of the uterus for the implantation of the egg. Estrogen regulates the development of breasts. Inhibin inhibits Follicle Stimulating Hormone (FSH) in females and in males it inhibits the development of sperm and androstenedione (androgen that is weaker than testosterone).

The testes produce androgens, testosterone and sperms.





Testes and Ovaries

Diseases related to endocrine system

- Diabetes insipidus
- Acromegaly
- Gigantism
- Syndrome of Inappropriate Antidiuretic Hormone Secretion (SIADH)
- Hyperthyroidism
- Hypothyroidism
- Cushing's syndrome
- Addison's disease

1.10 Reproductive System

The role of the male and female reproductive system is to procreate and to provide sexual gratification to the person.

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